

Home Task

Date: 19-06-2026

1. A company is building a Resume Screening System. The system receives raw resume text and must preprocess it to extract meaningful keywords for ranking candidates. You are required to build an NLP pipeline that performs the following steps: Tasks: 1. Tokenize the input text into words 2. Remove stopwords 3. Perform stemming 4. Perform lemmatization 5. Apply POS tagging 6. Compute frequency distribution of remaining words

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer

from nltk import pos_tag, FreqDist

nltk.download('punkt_tab')

nltk.download('stopwords')

nltk.download('wordnet')

nltk.download('averaged_perceptron_tagger')

nltk.download('omw-1.4')

resume_text="""

John Doe is a software engineer with 5 years of experience in Python,

Machine Learning, Data Analysis, SQL, and Web Development.

He has worked on AI projects and developed scalable applications.

"""

tokens=word_tokenize(resume_text)

print("Tokens:")

print(tokens)

stop_words=set(stopwords.words('english'))

filtered_tokens=[word.lower() for word in tokens if word.isalnum() and word.lower() not in stop_words]

print("\nAfter Stopword Removal:")

print(filtered_tokens)

stemmer=PorterStemmer()

stemmed_words=[stemmer.stem(word) for word in filtered_tokens]
```

```
print("\nStemmed Words:")
print(stemmed_words)
lemmatizer=WordNetLemmatizer()
lemmatized_words=[lemmatizer.lemmatize(word) for word in filtered_tokens]
print("\nLemmatized Words:")
print(lemmatized_words)
pos_tags=pos_tag(lemmatized_words)
print("\nPOS Tags:")
print(pos_tags)
freq_dist=FreqDist(lemmatized_words)
print("\nFrequency Distribution:")
for word,freq in freq_dist.items():
    print(word,":",freq)
```